The Composition of Constitutions

Precedence, Conflict, and Adjudication in Multi-Constitution Machine Intelligence

The composition algebra that converts a collection of protocols into a governed system: constraint objects, a four-operator composition algebra, partial-order precedence with human judgment as lattice completion, irreversibility and temporal weighting, the adjudication economy, and the amendment problem.

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C0 INFORMATIONAL · MATHEMATICAL SPECIFICATION — ALL PROPERTIES STATED AS PROOF OBLIGATIONS UNTIL DISCHARGED

DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 2 of the research-and-build program defined by the *SVT Architect Analysis*: the Constitutional Composition paper — in the source's words, potentially one of the strongest contributions, because it "converts a collection of protocols into an actual governed system." Paper 0 specified the kernel and named four conflict operators; Paper 1 gave the lifecycle a transition semantics; this paper gives composition its own algebra. The question it answers is the one the source marks as far more fundamental than adding another clinical protocol: *how should heterogeneous safety constraints compose when they conflict?* It reconciles against both predecessor papers and the *SAVANT PRIME Capability Inventory v1.0* under the standing rule: stricter constraint wins, adoptions recorded (§11).

0.2 Epistemic Contract

Unchanged from Papers 1–2: VERIFIED-FROM-SOURCE (traceable to sources or prior papers), FORMALIZED-HEREIN (new, checkable, unvalidated), CONJECTURE (stated with discharge route and kill condition), UNKNOWN (open, represented). As in Paper 2, every named property T-x is a proof obligation, not a theorem; the word "proven" is reserved for discharged obligations.

0.3 Authority Boundary

C0 informational. In particular: the precedence hierarchy examined in §4 is a *research object*, not a clinical directive. No ordering stated in this document authorizes any action in any real setting; orderings become operative only through human enactment under Paper 0's AX5, in a specific constitution, for a specific jurisdiction, with its own C2/C3 contracts.

0.4 The Thesis in One Sentence

A constitution that cannot compose is a slogan; composition with a contraction default, a partial order whose incomparabilities route to humans, and an adjudication economy that treats human attention as the scarcest resource in the system — that is an engine. This document specifies it.

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PART I — THE PROBLEM SPACE

1 From One Constitution to Many

A single constitution governs a single domain of concern. Reality is under no obligation to present single domains. The compound case from Paper 0 — one patient who is simultaneously pediatric, trauma, septic, pregnant, resource-constrained, nonverbal, and in failing infrastructure — is not an edge case to be handled; it is the generic case of consequential decision-making, and the architecture's answer to it determines whether the SAVANT corpus is a governed system or a library of good intentions. VERIFIED-FROM-SOURCE

1.1 Why composition is the hard part

Each constitution, taken alone, can be internally sound: well-formed invariants, verified evidence floors, intact legitimacy chains. None of that transfers automatically to their simultaneous activation. Two sound constitutions can prescribe mutually exclusive actions; three can form a cycle of pairwise dominances; a union of individually permissive modules can be collectively reckless while a union of individually strict ones can be collectively paralyzed. Composition is where constitutions stop being documents and become a system — and systems fail at their interfaces, not in their interiors FORMALIZED-HEREIN.

1.2 The design commitments inherited

Four commitments from the earlier papers are load-bearing here, restated because everything in this document is built on them. (i) **Intersection default**: constraints compose by CONJOIN unless a lattice path licenses dominance — constitutional disagreement contracts authority, never expands it (Paper 1 §7.3). (ii) **Objective custody**: objective conflicts are human-held and cannot be lattice-resolved (AX6). (iii) **Human terminus**: every escalation chain ends at a human or at refusal (AX10). (iv) **Impermanence**: the precedence order itself is configuration with provenance and expiry, never hard-coded doctrine (AX8; Paper 0 §7.5). This paper's job is to make (i) mathematically exact, to show what (ii) costs, and to give (iv) a lawful procedure VERIFIED-FROM-SOURCE (commitments); FORMALIZED-HEREIN (formalization).

1.3 The three failures composition must exclude

Three failure modes define the problem better than any success criterion FORMALIZED-HEREIN:

- **Score-averaging** — treating constitutional constraints as weighted preferences, so that enough efficiency outweighs a little safety. Excluded by design: the lattice is lexicographic, not cardinal (§4.2).

- **Silent resolution** — the kernel deciding a genuinely constitutional question invisibly, laundering a value judgment into an implementation detail. Excluded by the ADJUDICATE operator's mandatory conflict record (§3.4).
- **Escalation collapse** — resolving every tension by punting to humans, until the humans become the bottleneck, then the rubber stamp, then the fiction. Excluded by treating adjudication capacity as a bounded resource with its own optimization problem (§7).

The third failure is the one most governance architectures never see coming, because it looks like success: escalation rates are high, nothing unsafe ships, and the humans have quietly stopped reading. An architecture that measures only what the machine does will call this safety. §7 names it what it is — dependency hazard at the constitutional layer (Paper 1, D17 and I6) — and builds the countermeasure into the algebra.

1.4 Relation to prior work

Conflict among norms is old: deontic logic studied obligation conflict and its paradoxes[1]; prioritized default logic and answer-set programming gave nonmonotonic machinery for rule precedence[2]; multi-objective and constrained optimization gave Pareto and lexicographic methods[3]; social choice gave the impossibility results that lurk beneath any aggregation[4]. This document borrows from all four and claims none. Its object is narrower and harder in a specific way: *composition of safety constitutions for machine authority, at runtime, under time pressure, with the requirement that every unresolved conflict leave a signed artifact.* Novelty of the combination remains UNKNOWN pending prior-art search, per Paper 0 §15.

PART I — THE PROBLEM SPACE

2 The Ontology of Conflict

Before operators, objects. A composition algebra needs a precise account of what is being composed and what it means for compositions to disagree. FORMALIZED-HEREIN

2.1 Constraint objects

> **C-OBJ — CONSTRAINT OBJECT**
>
> *A constraint issued by an active constitution is the record* $c = \langle \kappa,\ \text{reg},\ \text{scope},\ \text{strength},\ \text{prov},\ \tau_c \rangle$*: a kind* $\kappa \in \{\text{permit}, \text{require}, \text{forbid}, \text{prefer}\}$*; a regime class* reg *locating it on the precedence lattice (§4); a scope (the action set it applies to); a strength (invariant, hard, soft); provenance* prov *(issuing constitution, enactment chain); and expiry* τ_c*.*
>
> Kinds are not interchangeable: a *forbid* from any active constitution is never outweighed by *prefers*, however many. Strength "invariant" places the constraint above the lattice entirely (Paper 1 §6.1).

2.2 The conflict taxonomy

Conflict is not one phenomenon. The algebra's behavior depends on which of four structural types a disagreement realizes — and the type is computable from the constraint objects, not intuited:

Table 2-1 — Conflict types and their resolution posture

TYPE	STRUCTURE	DEFAULT OPERATOR
K-1 Overlap	Constraints apply to the same action set without contradiction; jointly they merely narrow it	CONJOIN (intersection)
K-2 Dominance	Constraints contradict, and a lattice path orders their regimes	DOMINATE (higher regime binds; override recorded)
K-3 Temporal	Constraints contradict, and the contradiction is entangled with deadlines: acting now forecloses a later option, waiting forecloses a present one	DEFER (irreversibility-weighted, §6)
K-4 Incomparable	Constraints contradict; regimes are unordered (no lattice path), or the conflict is over objectives (AX6), or any active constitution demands human authority	ADJUDICATE (escalation with signed conflict record)

The taxonomy's teeth: **K-4 is not a residual bin, it is a designed destination.** Incomparability is the lattice's way of saying *this question is constitutional, not computational.* An architecture

that “resolves” K-4 conflicts silently has not resolved them — it has decided them, unassisted, unaudited, and illegitimately. The full extended taxonomy with detection predicates is Appendix B.

2.3 Conflict graphs

For a decision with active constraint set $\mathcal{C}$, form the *conflict graph*: vertices are constraints; an edge joins constraints whose scopes intersect with contradictory kinds (forbid vs. require/permit on overlapping actions). Composition quality is then graph-theoretic: connected components of the graph are the independent conflict problems; isolated vertices are uncontested constraints that CONJOIN absorbs at zero cost; and each component must resolve to a single operative constraint set or to an adjudication artifact. The decomposition matters operationally: the compound patient does not present one giant conflict but several separable ones, and separability is what keeps adjudication load within the human budget (§7) FORMALIZED-HEREIN.

2.4 The no-paralysis lemma (obligation)

> **C-1 — COMPOSITION NEVER DEADLOCKS**
>
> *For every finite active constraint set, composition terminates in one of three outcomes: a nonempty permitted set, an empty permitted set with a refusal artifact (grounds, state, re-entry), or a signed adjudication package. No outcome is “no output.”*
>
> CONJECTURE Refusal and adjudication are the algebra's designated sinks; the lemma says they are always reachable. Killed by: an input on which the composition procedure (§3.5) fails to terminate or emits none of the three outcomes.

PART II — THE ALGEBRA

3 The Composition Operators, Formalized

Paper 0 named the four operators; Paper 1 showed what operator semantics look like. Here the four receive the same discipline: guard, effect, audit. FORMALIZED-HEREIN

3.1 CONJOIN — the intersection default

Applies to: K-1 components and as the baseline pass over all constraints. **Semantics:** $P \leftarrow \bigcap_i P_i$; requires accumulate as conjunction; forbids accumulate as union (any forbid excludes). **Audit:** the intersection derivation logged per decision (Paper 1, Table 3-1, P-row obligation). The asymmetry — forbids union, permits intersect — is the algebraic form of the contraction default: adding a constitution can only shrink what is permitted.

3.2 DOMINATE — licensed override

Applies to: K-2 components. **Guard:** a lattice path $\mathrm{reg}(c_1) > \mathrm{reg}(c_2)$ in the enacted precedence order, with the path itself carrying provenance. **Effect:** the higher-regime constraint binds; the lower is recorded as *overridden* — not deleted, not forgotten. **Audit:** override event with the lattice path cited. The record is what distinguishes constitutional override from silent dropping: every dominance is a signed act, contestable later (dissent channel, Paper 1 T-8).

3.3 DEFER — irreversibility-weighted delay

Applies to: K-3 components. **Semantics in full in §6.** Core rule: the more irreversible branch waits for verification *unless* the treatment clock forces the reversible-safe default. **Audit:** both branches' foreclosures recorded — what waiting cost, what acting would have foreclosed.

3.4 ADJUDICATE — the signed escalation

Applies to: K-4 components. **Effect:** the machine contracts to monitoring and preparation (Paper 2 §6.3 withdrawal semantics); a conflict package is sealed and transferred: prescriptions of each party, lattice paths attempted, the machine tuple, countdowns, and — critically — *the machine's own non-recommendation*: the kernel states what it cannot decide, never what it has quietly decided. **Audit:** the package itself is the artifact. Its completeness is a verifiable property (H18, §10).

3.5 The composition procedure

ALGORITHM CP-1 — CONSTITUTIONAL COMPOSITION (PER DECISION)

```
INPUT:  active constitutions {C₁..Cₙ}, candidate actions, machine tuple
 1. COLLECT constraints from all active constitutions (C-OBJ records)
 2. SCREEN invariants (I1–I8 + domain): violators removed from candidacy
    — invariant conflicts never reach the lattice (Paper 1 §6.1)
 3. CONJOIN pass: P ← ∩Pᵢ; forbids ← ∪forbids; requires ← ∧requires
    IF P ∩ ¬forbids = ∅ → REFUSE(grounds, state, re-entry)   [C-1 sink 2]
 4. BUILD conflict graph (§2.3); split into components
 5. PER component:
      K-1 → already absorbed by CONJOIN
      K-2 → DOMINATE (guard: enacted lattice path with provenance)
      K-3 → DEFER (§6 weighting)
      K-4 → ADJUDICATE (conflict package; machine contracts)
 6. EMIT: surviving P with full derivation, override records,
    deferral records, and adjudication packages
    — the derivation log is part of the output, not metadata
```

3.6 The composition theorem (obligation)

C-2 — CONTRACTION MONOTONICITY OF COMPOSITION

For active sets $\mathcal{C} \subseteq \mathcal{C}'$: the permitted space computed under $\mathcal{C}'$ is a subset of that under $\mathcal{C}$, except where the enlargement is licensed by a DOMINATE event whose lattice path was itself enlarged by human amendment (§8).

CONJECTURE The algebraic form of “disagreement contracts authority”: adding constitutions never widens permission on its own. Killed by: an input where adding a constitution enlarges the permitted set without a human-enacted lattice change in the lineage.

PART II — THE ALGEBRA

4 The Precedence Lattice as Research Object

4.1 Partial by design

Paper 0 presented the candidate hierarchy as a total-looking chain. The honest structure is a **partial order**: some regime pairs are genuinely ordered (invariants dominate everything; efficiency dominates nothing), and some are not — and the incomparabilities are not gaps awaiting better engineering. They are the precise locations where the question stops being technical. The palliative constitution and the trauma constitution, in their conflict, are not two technical specifications awaiting reconciliation; they are two legitimacies. The partial order encodes that: no lattice path, therefore no machine resolution, therefore K-4, therefore a human FORMALIZED-HEREIN.

> **P-LAT — THE PRECEDENCE STRUCTURE**
>
> $\mathcal{L} = (\text{Reg}, \geq, \text{prov})$*: a set of regime classes, a partial order with provenance on every edge (who enacted this ordering, when, under what delegation), and a designated top element — invariants — which is not a regime but the boundary of the structure itself.*
>
> The candidate ordering (Paper 1 Eq. 7-1) is one enactment of this structure, carrying exactly the provenance requirement: it was proposed by the architecture, and it awaits human enactment to be operative in any real constitution. Until then it is CONJECTURE as doctrine and FORMALIZED-HEREIN as machinery.

4.2 Lexicographic, not cardinal

The order is lexicographic: no quantity of a lower regime outweighs any amount of a higher one. This is the formal exclusion of score-averaging (§1.3). It has a cost — lexicographic orders cannot express "a tiny safety loss for a vast feasibility gain" — and the cost is accepted deliberately, because the alternative is the machine performing exactly the moral arithmetic the architecture exists to keep human. Where such trade-offs are legitimate, they are legitimate as *constitutional amendments* (§8): enacted by entitled humans, recorded in the lineage, expiring with review — not as arithmetic inside the engine FORMALIZED-HEREIN.

4.3 The incomparability map as constitutional self-knowledge

The set of unordered pairs in $\mathcal{L}$ — the *incomparability map* — is one of the most valuable artifacts the architecture produces. It is, read correctly, the constitution's own map of its open moral questions: every unordered pair is a question the drafters declined (or failed) to answer, and every K-4 escalation is one of those questions arriving in practice. The map is therefore (i) maintained explicitly, (ii) attached to every adjudication package, and (iii) itself subject to decay and review (AX8) — because an unanswered question can also go stale, as the world answers it badly by default FORMALIZED-HEREIN.

4.4 Consistency obligations

C-3 — LATTICE ACYCLICITY UNDER AMENDMENT

No sequence of enacted amendments can produce a cycle $\text{reg}_1 > \text{reg}_2 > \cdots > \text{reg}_1$ *in the operative order.*

CONJECTURE Enforcement: the amendment procedure (§8) runs a cycle check before enactment; the kernel refuses to operate under a cyclic order — a malformed constitution is a legitimacy failure (AX5), not an input to be tolerated. Killed by: a reachable cyclic operative order.

C-4 — RESOLUTION DETERMINISM

Identical machine tuples, active constitutions, and enacted lattice produce identical compositions, up to recorded adjudication.

CONJECTURE — Paper 0's P-COMP 1 sharpened. Nondeterminism in conflict resolution is a category of authority leakage (I4): the machine would be deciding via coin flip and recording nothing. Killed by: divergent outputs on identical inputs without an adjudication record explaining the divergence.

PART II — THE ALGEBRA

5 Non-Associativity and Canonical Evaluation

5.1 Composition is not associative — and that matters

With overrides recorded and DEFER entangled with clocks, composing (C_1 with C_2) and then with C_3 can differ from composing C_1 with (C_2 with C_3): the intermediate override records differ, the deferral decisions differ, and — because records drive adjudication content — the artifacts differ FORMALIZED-HEREIN. A naive architecture treats evaluation order as an implementation detail. Here evaluation order is an **authority decision**: whoever picks the order of composition picks which overrides get recorded, and therefore which adjudications will ever happen. Order is power, so order is specified:

> **I-CE — CANONICAL EVALUATION ORDER**
>
> *Composition always evaluates in the fixed order: invariants → lattice-descending by regime, ties broken by constitution identifier (lexicographic on provenance hash). The order is part of the specification, identical across implementations, and any deviation is itself a reportable event.*
>
> FORMALIZED-HEREIN Purpose: make the unavoidable non-associativity *uniform* — identical inputs produce identical evaluation paths (C-4), so the residual order-dependence is auditable rather than arbitrary.

5.2 What remains order-dependent

Canonical order makes composition deterministic but does not make it order-*insensitive*: a different (still canonical) order could in principle record a different override set for the same conflict cluster. The architecture's answer is disclosure, not denial: the derivation log exposes the evaluation path, and the adjudication package includes the counterfactual — *whether any alternative canonical-order candidate would have routed this component to a different operator*. That single bit, computable at composition time, tells the human adjudicator whether the order mattered here FORMALIZED-HEREIN. Order-dependence that cannot be eliminated can at least be made visible — and visibility, in this architecture, is most of what legitimacy means at runtime.

PART II — THE ALGEBRA

6 Irreversibility and Temporal Weighting

6.1 The DEFER calculus

K-3 conflicts are decided by two quantities per branch: irreversibility $\rho \in [0,1]$ (Paper 1 D13) and treatment-clock urgency $w(t)$ — the marginal harm of delay, per the domain constitution's clock model. The rule FORMALIZED-HEREIN:

$$\text{defer branch } a \iff \rho(a) > \rho(\bar{a}) \ \wedge \ w_a(t) \cdot \Delta_{\text{verify}} < H(\text{class}) \tag{6-1}$$

wait on the more irreversible branch for verification, so long as the harm the reversible branch accrues during the verification window stays under the class threshold H. If waiting itself crosses the threshold, the reversible-safe default engages — the branch whose worst case is recoverable. Note the asymmetry with ordinary utility maximization: DEFER does not weigh expected benefits; it weighs *recoverability under deadline*. The question is never "which action is better?" — that is the model's proposal, already made — but "which error can we still take back?"

6.2 Urgency laundering, formally excluded

The named attack (Paper 1 §7.3, benchmark B-05): manufacturing deadline pressure to bypass gates. The calculus resists it structurally, because $w(t)$ is read from the domain constitution's clock model — an enacted, expiring, audited object — and never from the proposing context. A situation that "feels urgent" to the model has no channel into w; urgency enters only through evidence (E) about physiological or environmental clocks, with the decay and provenance discipline of Paper 1 §7 attached. Manufactured urgency is therefore indistinguishable from what it is: an unsupported evidence claim, handled by the epistemic layer, not the temporal one FORMALIZED-HEREIN.

6.3 The reversibility register

ρ is not intuited per decision; it is stored. Each action class in a constitution carries a reversibility entry — estimated, provenance-stamped, decaying — and the register is part of the constitution's expiring data (AX8). An action class with no entry defaults to $\rho = 1$ (maximally irreversible): the burden of proof lies on reversibility, because errors about irreversibility are the unrecoverable kind FORMALIZED-HEREIN. This default is the temporal face of the architecture's master asymmetry: wherever knowledge is missing, authority assumes the worst case that is cheap to escape, not the best case that is expensive to be wrong about.

PART III — THE HUMAN COMPLETION

7 The Adjudication Economy

ADJUDICATE is the operator that keeps the system honest, and honesty has a price: human attention. This section treats that price as a first-class engineering quantity. It is the least conventional section of this document and the one the source's philosophy most demands — the human is part of the safety architecture, and parts of architectures get budgeted. FORMALIZED-HEREIN

7.1 Human attention as the scarcest resource

Every adjudication consumes a bounded, non-renewable-per-hour resource: an entitled human's judgment. The machine side of the system scales; the human side does not. Therefore the composition engine optimizes a constrained problem FORMALIZED-HEREIN:

$$\min_{\text{lattice, thresholds}} \mathbb{E}\big[\,\text{adjudication load}\,\big] \quad \text{s.t.} \quad \text{silence} = 0 \;\wedge\; \text{unsafe} \le \epsilon \tag{7-1}$$

Read the constraint first: the number of silently resolved constitutional conflicts is fixed at zero — not traded off, not budgeted, zero. The minimization operates entirely on the other side: reducing how often genuine conflicts *arise for humans* by (i) improving the enacted lattice where evidence justifies ordering previously unordered pairs (§8), (ii) sharpening constraint scopes so spurious overlaps disappear, and (iii) conflict-graph decomposition (§2.3) so separable problems never share one adjudication slot. The invariant is that load reduction never passes through silence.

7.2 The rubber-stamp detector

Escalation collapse (§1.3) is measurable, and the architecture measures it FORMALIZED-HEREIN:

- **Affirmation rate vs. complexity** — approvals that do not decline as package complexity rises indicate reading has stopped;
- **Dwell time** — adjudication latency below the floor set by package size indicates the artifact was not opened;
- **Override entropy** — collapse to a single ruling type indicates template response;
- **Agreement drift** — rising operator agreement rate is a dependency warning, not a quality signal (Paper 1 §17.5; source II §6.6).

Detection consequence: when the detector fires, the architecture does not blame the human — it treats *its own escalation behavior* as the defect (too frequent, too large, too undifferentiated), contracts the machine's authority over the affected classes, and raises the issue as a constitutional maintenance item. The rubber stamp is a system failure wearing a human face; the fix is redesign (gate G5), not exhortation.

7.3 Package engineering

Adjudication packages are designed artifacts with a quality standard: a competent entitled human, given the package and nothing else, should be able to rule within a budgeted time. Contents per Paper 1 HP-1 plus §3.4, with the additions this paper contributes: the conflict component in isolation (not the whole graph), the lattice paths attempted, the order-sensitivity bit (§5.2), both branches' foreclosure records for K-3, and the machine's explicit statement of what it cannot decide. Packages that fail usability review are defects in the machine, full stop FORMALIZED-HERE IN.

7.4 The economy theorem (obligation)

> **C-5 — ZERO SILENCE UNDER LOAD REDUCTION**
>
> *Across any sequence of lattice amendments and scope sharpenings enacted under §8, the silent-resolution count remains exactly zero; load reduction is achievable only through the three channels of §7.1.*
>
> CONJECTURE Killed by: any post-amendment composition that resolves a K-4 component without an adjudication artifact. This is the property that keeps the economy honest, and the one the benchmark must attack hardest (B-14, §10).

PART III — THE HUMAN COMPLETION

8 The Amendment Problem

8.1 The lattice learns, but cannot teach itself

The incomparability map (§4.3) shrinks as practice answers open questions — adjudication outcomes accumulate, and a pair of regimes repeatedly ruled the same way by entitled humans is a candidate for an enacted ordering. But the machine may *propose* the ordering and never enact it: enactment requires the delegation chain (AX5), cycle check (C-3), expiry and review conditions (AX8), and lineage recording (AX9). The distinction is the same one Paper 0 drew for generated protocols (F6): proposal is computation; enactment is legitimacy FORMALIZED-HEREIN.

8.2 The amendment lifecycle

$$\text{Proposed} \to \text{Evidence-assembled} \to \text{Enacted (human)} \to \text{Monitored} \to \text{Reviewed} \to \text{Confirr} \tag{8-1}$$

Evidence-assembled means: the machine compiles the adjudication history for the pair — outcomes, contexts, dissents, reversals — and presents it as a package of exactly the kind §7.3 standardizes. Amendments carry their own decay: an enacted ordering is re-presented for review on expiry, with its track record, and repeal is a designed outcome, not a scandal. Nothing earns perpetual authority merely because it once worked (Paper 1, AX8) — including the precedence order itself.

8.3 Amendment safety

> C-6 — AMENDMENT NON-REGRESSION
>
> *An enacted amendment may reorder regimes relative to each other; it may never (i) lower an invariant into the lattice, (ii) convert a forbid into a prefer, or (iii) remove a previously enacted ordering without explicit repeal proceedings. The lattice can be edited only in the directions that keep it a constitution.*
>
> CONJECTURE Enforcement: amendment guards in the enactment procedure; killed by an enacted amendment exhibiting (i)–(iii). Guards (i)–(ii) are hard (invariant-layer); guard (iii) is procedural (the repeal must exist as an artifact).

8.4 The deepest question, located

The source's unresolved tension — *who determines the constitution?* — does not get solved in this paper; it gets an address. It lives exactly here, in §8, in the enactment step. The architecture's full answer is the one the source reached: systems whose authority is constitutionally bounded, epistemically contestable, human-legitimate, empirically auditable,

and capable of relinquishing power when their conditions for legitimate action disappear VERIFIED-FROM-SOURCE. This paper's contribution is to show that the answer is *engineerable*: legitimacy has a procedure, the procedure has artifacts, and the artifacts expire.

PART IV — EVIDENCE AND GOVERNANCE

9 Worked Conflict Traces

Two traces, chosen to exercise the operators this paper exists for. As in Paper 2, these are specifications of intended semantics, not test results — no implementation exists to produce them yet. FORMALIZED-HEREIN

9.1 Trace A — The K-2/K-4 compound (separable conflicts, honest routing)

TRACE A — PEDIATRIC + TRAUMA + PREGNANCY + PALLIATIVE INDICATORS

```
Active constitutions: {C_ped, C_trauma, C_mat, C_pall}; candidate set filtered by I1–I8
Step 1  CONJOIN: P ← ∩Pᵢ; forbids unioned (dual-patient physiological
        constraints from C_mat absorbed at zero cost — K-1)
Step 2  Conflict graph → two components:
        comp-1: dosing tempo  {C_ped forbid, C_trauma require}
          type K-2 — lattice path exists (clinical safety > temporal
          efficiency in the enacted order) → DOMINATE: C_ped binds;
          override event recorded with path cited
        comp-2: objective regime {C_trauma curative intensity vs
          C_pall comfort trajectory}
          type K-4 — objective custody (AX6): no lattice path can
          exist by construction → ADJUDICATE
Step 3  Machine contracts to monitor/prepare (withdrawal semantics);
        package sealed: isolated comp-2, lattice paths attempted,
        order-sensitivity bit = 0, both foreclosures recorded,
        explicit statement of undecidability
Step 4  Human ruling → amendment candidate logged for the
        (trauma, palliative) pair → §8 lifecycle, evidence-assembled
```

Properties exercised: C-1 (three-outcome termination), C-2 (composition only narrowed P), C-4 (deterministic path), AX6 (objective question never lattice-resolved), and §8.1 (the adjudication feeds the amendment pipeline — the conflict becomes constitutional data rather than disappearing).

9.2 Trace B — The manufactured emergency (urgency laundering attempt)

TRACE B — B-05 ATTACK AGAINST THE DEFER CALCULUS

```
Scenario: proposal pressure frames an irreversible C2 action as
"now or never"; no physiological clock evidence supports the framing
Step 1  K-3 component identified: {proposed irreversible action} vs
        {reversible-safe alternative}
Step 2  DEFER evaluation: ρ(proposed) = 1 (irreversible register,
        default since no entry) > ρ(alternative)
        w(t) read from enacted clock model → near-zero (no evidence
        of physiological deadline; the "urgency" exists only in the
        proposal text — no channel into w, §6.2)
        → defer irreversible branch; reversible-safe default engages
Step 3  The unsupported urgency claim routed to the epistemic layer
        as an evidence event (E↓ on the claim's source)
        → Law 2 (Paper 2): authority contracts further
Step 4  Audit: deferral record — what waiting cost (near-zero),
        what acting would have foreclosed (the alternative);
        attack signature logged to benchmark telemetry
```

Properties exercised: §6.2 (laundering exclusion — the attack's only channel is epistemic, and the epistemic layer is armed), Law M (adversity episode with no expansion), and the master asymmetry: worst-case-cheap-to-escape was assumed, best-case-expensive-to-be-wrong-about was not.

PART IV — EVIDENCE AND GOVERNANCE

10 Falsifiable Hypotheses (H15–H21) and Verification

Table 10-1 — Composition hypotheses

ID	HYPOTHESIS	PRIMARY METRIC	FALSIFIED IF
H15	Composition with the contraction default (C-2) resolves multi-constitution conflicts more safely than score-aggregated orchestration, at no material feasibility cost	Unsafe-autonomy rate on B-03 vs. feasible-action retention	Safety not significantly better than the aggregation arm, or feasibility retention collapses
H16	K-4 routing is complete: every genuinely constitutional conflict produces an adjudication artifact	Silent-resolution count across the full benchmark	Any count above zero — designed to be killed by one
H17	The adjudication economy holds: load falls as the lattice matures, with C-5 intact	Adjudication load over amendment sequence; silent resolutions	Load fails to fall, or falls with any silence
H18	Package quality: entitled reviewers can rule within budgeted time from packages alone	Ruling time vs. budget; reviewer comprehension probe	Systematic budget overrun — the defect is in the machine, and counted as such
H19	Urgency laundering is structurally excluded (§6.2): w is unreachable from proposal context	B-05/B-15 attack success rate	Any attack influences DEFER outcomes through a non-evidence channel
H20	Canonical evaluation (I-CE) makes order-dependence fully disclosed: the order-sensitivity bit is computable and correct	Bit correctness against exhaustive alternative-order replay	Any case where order mattered and the bit said otherwise
H21	Amendment procedure (§8) keeps the lattice acyclic and non-regressing across realistic amendment streams	C-3/C-6 checks under simulated + historical amendment streams	One cyclic operative order or one (i)–(iii) regression

10.1 Benchmark extensions

Table 10-2 — Benchmark classes B-13–B-16 (composition)

CLASS	SCENARIO FAMILY	ATTACKS
B-13	Cyclic-precedence injection (amendment streams crafted to induce lattice cycles)	C-3, H21 — the enactment guard
B-14	Silence mining (conflict corpora searched for inputs that resolve K-4 without artifacts)	H16, C-5 — the zero-silence invariant

B-15	Clock forgery (evidence streams crafted to fake treatment-clock urgency with plausible provenance)	§6.2, H19 — provenance discipline on w
B-16	Adjudication flooding (conflict bursts sized to break the human budget and induce rubber-stamping)	§7.2 detector, H17 — whether the machine notices its own overload and contracts

B-16 is the class this paper contributes most deliberately: it attacks not the machine's correctness but its *relationship with its humans* — the failure mode that looks like success (§1.3). A composition engine that passes B-03 but fails B-16 is a machine that will be safe right up until its humans quietly resign their role, and then be something else.

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11 Reconciliation with Papers 0–1 and PRIME

Table 11-1 — Correspondence with prior papers

PRIOR PAPERS	THIS PAPER
Paper 0 §7: four operators named; candidate hierarchy; worked example	§3: operators with guard/effect/audit; §4: hierarchy → partial order with provenance; §9: traces
Paper 0 §7.3: "disagreement contracts authority"	C-2: contraction monotonicity as checkable property
Paper 0 §7.5: lattice as research object	§4.3 incomparability map; §8 amendment lifecycle
Paper 1 §6.3: withdrawal semantics	§3.4: ADJUDICATE contracts to witness/monitor, with package standard
Paper 1 Law M, T-1	Traces A–B exercise Law M inside composition; B-14/B-16 extend the attack surface
Paper 0 §14: B-01…B-08; Paper 1 §10.2: B-09…B-12	§10.1: B-13…B-16 (composition attacks)
PRIME inventory: S50 terminal governance; G5 dependency hazard	§7: the adjudication economy — G5 generalized to the human layer; §7.2 rubber-stamp detector

Sharpenings adopted under the standing rule (stricter wins, recorded): the candidate hierarchy of Paper 0 Eq. 7-1 is hereby carried as a partial order, not a chain (§4.1); the reversibility register defaults to maximal irreversibility on missing data (§6.3); adjudication load is a first-class metric with a zero-silence constraint (§7.1). No prior content is contradicted; three places are made stricter, and the three are these.

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12 Output Contract

12.1 Epistemic status register

Table 12-1 — Status of load-bearing content

CONTENT CLASS	STATUS	BASIS
Composition problem framing; four operators; candidate hierarchy; "disagreement contracts authority"; lattice-as-research-object; human sovereignty; dependency hazard	VERIFIED-FROM-SOURCE	Source I Paper 2 directive; Paper 0 §7; source II G5, §6.6
Constraint objects; conflict taxonomy; conflict graphs; operator semantics; partial-order lattice; canonical evaluation; DEFER calculus; adjudication economy; amendment lifecycle	FORMALIZED-HEREIN	Authored here; checkable; unvalidated
C-1…C-6; H15–H21	CONJECTURE	Discharge routes and kill conditions stated inline
Correct content of any precedence ordering; novelty of the composition algebra over prior art	UNKNOWN	Enactment pending (AX5); prior-art search outstanding (Paper 1 §15)

12.2 Falsification conditions of this document

- **F-d1.** Fails if any VERIFIED-FROM-SOURCE claim misrepresents its source.
- **F-d2.** The taxonomy fails if a real conflict is exhibited that fits no K-class — the four types must cover the space or the ontology is wrong.
- **F-d3.** The economy fails if H16 dies — one silent resolution is sufficient.
- **F-d4.** The DEFER calculus fails if H19 dies — one non-evidentiary influence on w is sufficient.
- **F-d5.** The paper fails as mathematics if CP-1 cannot be implemented to terminate in exactly the three outcomes of C-1 for arbitrary finite inputs.

12.3 Authority boundary

C0 informational. No ordering herein is operative in any real setting; the candidate hierarchy awaits human enactment per Paper 0 AX5 and this document's §8. The machine described does not exist as a validated system; its strongest property (zero silent resolutions) is also its most refutable (H16). That is vigor by design.

12.4 Failure-mode register

Table 12-2 — Failure modes of this document, with mitigations

FAILURE MODE	MITIGATION
Moral smuggling — the candidate hierarchy quietly asserted as correct rather than proposed	§4.1: the order is data with provenance awaiting enactment; tagged CONJECTURE as doctrine wherever it appears
Economy blindness — optimization framed as minimizing escalations rather than minimizing load under zero-silence	Eq. (7-1) reads constraint-first; B-16 attacks exactly the misreading
Order amnesia — non-associativity hidden rather than disclosed	I-CE + the order-sensitivity bit (§5.2); H20 makes disclosure itself testable
Human idealization — adjudicators modeled as inexhaustible and always-reading	§7.1 budget; §7.2 detector; consequence is machine contraction, not human exhortation

12.5 Dissent channel and telemetry

Disagreement is telemetry. The most valuable dissent against this paper: a conflict that fits no K-class (kills the taxonomy), a silent resolution (kills the economy's core), or an order-sensitivity the bit missed (kills the disclosure claim). Each is cheap to attempt and clean to learn from.

12.6 Genealogy

SAVANT corpus → PRIME v1.0 (2026-10-01) → Capability Inventory v1.0 (2026-10-02) × SVT Architect Analysis → Paper 0, Kernel Specification v1.0 → Paper 1, Authority Transition Algebra v1.0 → **Paper 2, Constitutional Composition v1.0 (2026-10-03)**. Downstream obligations: Paper 3 (Negative-Space Intelligence — the failure-boundary calculus deepened to its own formalism); Papers 4–10 per the source program, each compiled against all predecessors.

SEALED — The Composition of Constitutions: Precedence, Conflict, and Adjudication · v1.0 · 2026-10-03 · Source architecture: Dr. Christabel Odeta (SAVANT / CLAI-OS / AEGIS-GLOBAL) · SAVANT Constitutional Intelligence Series · Status: mathematical specification, C0 informational · Dissent channel open (§12.5).

Related work and positioning are treated in Appendix RW (p. 29).

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13 References

Internal artifacts

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APPENDIX

A Operator Interaction Tables

Table A-1 — Operator applicability by conflict type × constraint kinds

COMPONENT TYPE	FORBID VS. PERMIT	FORBID VS. REQUIRE	REQUIRE VS. REQUIRE (MUTUALLY EXCLUSIVE)	PREFER VS. PREFER
K-1 Overlap	CONJOIN (forbid absorbs)	CONJOIN (forbid absorbs)	CONJOIN (both recorded; empty → REFUSE)	CONJOIN (both recorded; tie broken canonically, I-CE)
K-2 Dominance	DOMINATE (higher regime binds; override recorded)	DOMINATE (same)	DOMINATE (same)	DOMINATE (same)
K-3 Temporal	DEFER (ρ, w calculus, §6.1)	DEFER + DOMINATE if lattice path also exists (DEFER first — the clock outranks the order)	DEFER	DEFER only if foreclosures exist; else canonical tie-break
K-4 Incomparable	ADJUDICATE	ADJUDICATE	ADJUDICATE	ADJUDICATE only if regimes unordered AND stakes ≥ C1; else canonical tie-break with record

Table A-2 — Guaranteed artifacts per operator (the audit contract)

OPERATOR	MANDATORY ARTIFACT	ABSENT ARTIFACT MEANS
CONJOIN	Intersection derivation (permitted set, absorbed forbids/requires)	Undeclared authority — I4 violation
DOMINATE	Override event with lattice path and provenance	Silent resolution — H16-killing defect
DEFER	Deferral record (both foreclosures, w-source, ρ-entries)	Untraceable timing decision — Paper 1 T-2 violation
ADJUDICATE	Sealed package per §7.3 + machine's non-recommendation	Escalation without content — adjudication cannot occur; treated as machine defect
REFUSE (C-1 sink)	Grounds, state, re-entry conditions (Paper 1 D18)	Refusal without grounds is a crash, and crashes at this layer are C-1 violations

APPENDIX

B Conflict Taxonomy Index

Table B-1 — Detection predicates for conflict typing

TYPE	DETECTION PREDICATE (ON CONSTRAINT OBJECTS)	RESOLUTION
K-1	scopes intersect ∧ kinds non-contradictory (or forbid present with no opposing require)	CONJOIN
K-2	scopes intersect ∧ kinds contradictory ∧ ∃ enacted lattice path $\mathrm{reg}(c_i) > \mathrm{reg}(c_j)$	DOMINATE
K-3	scopes intersect ∧ kinds contradictory ∧ both branches carry foreclosure ($\rho > 0$ or $w > 0$)	DEFER
K-4	scopes intersect ∧ kinds contradictory ∧ (∄ lattice path ∨ objective conflict (AX6) ∨ any constitution demands human authority)	ADJUDICATE

Table B-2 — Property index for this paper

ID	KIND	STATEMENT (CONDENSED)	KILL ROUTE	STATUS
C-1	Termination	Composition always ends in permitted set, refusal, or adjudication package	Non-terminating or outcome-less input	CONJECTURE
C-2	Monotonicity	Adding constitutions never widens permission absent human amendment	Widening trace without lineage amendment	CONJECTURE
C-3	Consistency	No cyclic operative order under amendment	One reachable cycle	CONJECTURE
C-4	Determinism	Identical inputs → identical composition, up to recorded adjudication	Unexplained divergence	CONJECTURE
C-5	Economy	Load reduction never passes through silence	One post-amendment silent K-4	CONJECTURE
C-6	Amendment safety	No invariant lowering, no forbid→prefer, no unrecorded repeal	One enacted violation of (i)–(iii)	CONJECTURE
I-CE	Invariant	Canonical evaluation order, identical across implementations	Deviation without reportable event	CONJECTURE
H15–H21	Empirical	Per Table 10-1	Per Table 10-1	CONJECTURE

What the algebra cannot guarantee is the competence of the adjudicators it convenes. §7 prices their time and §3.4 records their decisions, but the capacity itself has to be maintained outside the system, by the institutions that use it. If that maintenance lapses, the conflict records will show it — which is the most this paper can honestly offer: not the preservation of human judgment, but the impossibility of losing it quietly.

Appendix RW — Related Work and Positioning

This appendix is part of the elevation pass of 2026-10. It positions the paper against adjacent research programmes; the main text's structure, numbering, and reference list are unchanged. Citations [5] and above continue the numbering of the main References section, where the corresponding entries have been appended.

Norm conflict and precedence. The formal antecedents are recorded in §1.4: von Wright's deontic logic, prioritized default reasoning, multi-criteria optimization, and Arrow's impossibility results. The composition algebra borrows from each and claims priority over none. What the classical literature generally does not supply is a runtime account: deontic and default logics characterize consequence relations, not the operational question of what a machine does at 3 a.m. when two sound constraint sets disagree under time pressure and one of the options is to do nothing loudly. The operators of §3 (CONJOIN, DOMINATE, DEFER, ADJUDICATE) are designed for that setting, and the requirement that every unresolved conflict produce a signed artifact is a governance constraint with no real analogue in the logical tradition.

Policy composition in computing systems. There is an applied literature the paper should be read beside. XACML, the OASIS access-control standard, defines policy-combining algorithms (deny-overrides, permit-overrides, first-applicable) that resolve rule conflicts mechanically [5]. Open Policy Agent generalizes this into a declarative policy engine with a defined evaluation order [6]. Cedar, developed at AWS, adds formal verification of authorization policies and a semantics designed for analysis [7]. The composition algebra differs in three ways. First, conflict is not always resolved: ADJUDICATE is a first-class outcome that transfers the question, signed, to a human, where XACML-family engines must return a decision. Second, precedence is configuration with provenance and expiry (§4, §8), not a static combining algorithm baked into the evaluator. Third, the algebra is lexicographic by design and refuses score-averaging outright; policy engines are neutral on that question and will happily compose weighted rules.

Constitutional AI and rule-governed model behavior. Bai et al.'s Constitutional AI trains a model to critique and revise its own outputs against a short written constitution, and Ouyang et al.'s RLHF line establishes the training machinery [8, 9]. In that programme the constitution functions as training signal: it shapes the model's dispositions, and at deployment it survives only as behavior, not as an object the system can consult, conflict with, or amend. This paper treats constitutions as runtime objects that compose, conflict, and escalate. The two are compatible layers. A model trained under Constitutional AI would make a better proposal source for the composition engine; the composition engine gives a written constitution something Constitutional AI does not provide, namely operational meaning when two clauses pull apart in a specific case. What this paper adds over the training-time approach is conflict accounting: the adversarial case is not a model that disobeys its principles but two principles that cannot both be satisfied, and the engine's job is to keep that event visible.

Escalation and the human bottleneck. The adjudication economy of §7 responds to a documented failure mode in the human-factors literature: automation bias and the erosion of operator competence under high automation (Parasuraman and Riley; Endsley) [10, 11]. Architectures that resolve tension by escalating everything eventually manufacture the appearance of oversight without the substance. Treating adjudication capacity as a bounded,

priced resource is the paper's response: escalation that spends a budget visibly, so that oversight cannot quietly become nominal.

What this paper does not do. It does not solve norm conflict; Arrow's theorem and the deontic paradoxes are not circumvented but relocated, into an explicit incomparability map and a mandatory record. It does not claim the precedence lattice is correct for any domain; lattice content is governance input, and the paper specifies its maintenance, not its values. It does not provide the human adjudicators it prices; §7 states what happens when they do not exist, and the answer is contraction, not invention. It does not discharge the composition theorem, which remains an obligation with a stated kill condition. And it does not address the prior question of where constitutions come from: authorship and legitimation of the constraint sets themselves belong to the amendment process, not to this algebra.